

Milk Distribution SaaS - Enterprise PRD / BRD v6

Detailed Product Requirements & Business Requirements Document

1. Product Vision

Build a scalable SaaS platform for milk distributors, dairy businesses and consumers across India.

2. Business Objectives

Digitization, subscription automation, billing automation, distributor growth and customer retention.

3. Stakeholders

Platform Owner, Super Admin, Distributor, Delivery Staff (future), Customer.

4. User Personas

Detailed personas for distributors and customers including operational pain points.

5. Admin Modules

Verification Queue, Distributor Lifecycle, Customer Monitoring, Subscriptions, Deliveries, Billing, Notifications, Settings.

6. Distributor Modules

Dashboard, Customers, Products, Pricing, Delivery Slots, Subscription Management, Billing, Payments, Reports.

7. Customer Modules

Distributor Discovery, Subscription Setup, Pause Requests, Extra Milk Requests, Billing, Notifications.

8. Product Catalog

Milk, Curd, Paneer, Eggs, Lassi, Ghee, Butter, Khoya with configurable pricing.

9. Subscription Engine

Daily, Alternate Day, Weekdays, Weekly, Monthly, future Custom Schedule.

10. Delivery Engine

Generate daily delivery lists, support pauses, extra quantities and future route planning.

11. Billing Engine

Compute invoices based on delivered quantities, pricing and adjustments.

12. Notification Engine

In-app notifications with future SMS, WhatsApp and Email integration.

13. RBAC Matrix

Role-based access rules for every major feature and module.

14. Database Design

Users, DistributorProfiles, CustomerProfiles, Products, Pricing, DeliverySlots, Subscriptions, Deliveries, Bills, Payments.

15. API Inventory

Authentication APIs, Distributor APIs, Customer APIs, Billing APIs, Notification APIs and Admin APIs.

16. Security Requirements

JWT, Refresh Tokens, Validation, Audit Trails, Approval Workflows.

17. Non-Functional Requirements

Performance, scalability, availability, observability, backups and monitoring.

18. Testing Strategy

Unit, Integration, E2E, UAT, Load Testing and Regression Testing.

19. Release Strategy

Alpha → Beta → Pilot → Production rollout.

20. Future Roadmap

Wallets, Payments, Mobile Apps, Route Optimization, IoT, Analytics, Loyalty Systems.

Recommended Next Documentation (v7)

A screen-by-screen specification document containing wireframes, acceptance criteria, user stories, database ERD, API contracts, QA test cases and implementation roadmap.